Supplementary material

Reversible graph representation of ECG dynamics
for detecting functional state changes

Table S1: Variability of interval and amplitude series across test phases (paired Wilcoxon test).

Series	Index	Rest	Tilt 45°	p	n
RR intervals	SDNN, ms	58.9 ± 23.8	65.7 ± 33.3	0.105	20
	RMSSD, ms	59.5 ± 33.0	44.8 ± 53.0	0.002	20
	pNN50, %	34.2 ± 21.7	12.7 ± 15.5	< 0.001	20
	CV, %	6.2 ± 2.4	7.9 ± 4.0	0.006	20
PR intervals	SD, ms	3.8 ± 2.4	3.6 ± 2.6	0.579	17
	RMSSD, ms	4.4 ± 3.4	3.5 ± 3.3	0.015	17
	CV, %	2.5 ± 1.4	2.4 ± 1.6	0.927	17
RT intervals	SD, ms	3.6 ± 4.9	7.8 ± 9.9	0.001	20
	RMSSD, ms	3.8 ± 5.9	6.6 ± 15.4	0.571	20
	CV, %	1.4 ± 1.6	3.5 ± 5.1	< 0.001	20
R amplitude	mean, mV	1.227 ± 0.476	1.236 ± 0.477	0.985	20
	SD, mV	0.063 ± 0.031	0.083 ± 0.038	0.006	20
P amplitude	mean, mV	0.043 ± 0.041	0.063 ± 0.055	0.064	17
	SD, mV	0.067 ± 0.023	0.076 ± 0.023	0.109	17
T amplitude	mean, mV	0.425 ± 0.207	0.340 ± 0.199	< 0.001	20
	SD, mV	0.072 ± 0.023	0.081 ± 0.029	0.189	20

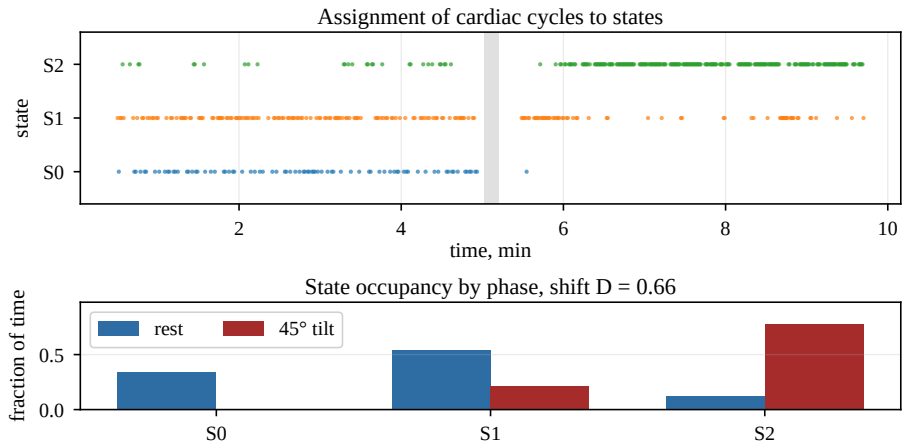


Figure S1: Assignment of cardiac cycles to states over time (top) and state occupancy by test phase (bottom) for one subject (recording 0011); the grey band marks the rise of the table.

Table S2: Within-phase null control by subject: state vector with the cycle duration (columns 2–3), without it (columns 4–5) and fixed-window features independent of the RR interval (columns 6–7). Null is the larger of the two within-phase values; effect is the shift between rest B and tilt A.

Subject	Null	Effect	Null	Effect	Null	Effect
0011	0.10	0.56	0.11	0.54	0.29	0.71
0012	0.23	0.35	0.24	0.40	0.30	0.70
0013	0.06	0.93	0.06	0.93	0.15	0.91
0014	0.11	0.76	0.11	0.76	0.01	1.00
0015	0.43	0.13	0.42	0.11	0.03	0.97
0016	0.15	0.76	0.07	0.87	0.02	0.97
0017	0.23	0.41	0.23	0.41	0.02	0.52
0018	0.75	0.25	0.74	0.26	0.20	0.80
0019	0.26	0.72	0.26	0.72	0.34	0.66
0020	0.38	0.56	0.38	0.55	0.43	0.69
0021	0.46	0.24	0.57	0.23	0.21	0.42
0022	0.08	0.22	0.06	0.22	0.46	0.54
0023	0.24	0.72	0.25	0.71	0.00	1.00
0024	0.25	0.66	0.25	0.90	0.17	0.81
0025	0.12	0.65	0.12	0.67	0.04	0.93
0026	0.24	0.15	0.24	0.11	0.08	0.18
0027	0.17	0.08	0.20	0.11	0.24	0.20
0028	0.02	0.98	0.01	0.99	0.03	0.98
0029	0.15	0.78	0.15	0.78	0.15	0.77
0030	0.03	0.15	0.04	0.59	0.08	0.14

Table S3: Graph characteristics for recordings with different rhythm disturbances.

Recording	Database section	States	HR of dominant state, bpm	Share of time	Stability	Transition entropy
07_VORO	bradycardia, narrow QRS	2	29	55%	0.91	0.43
10_MITI	bradycardia, narrow QRS	2	43	70%	0.65	0.86
01_GUSA	sudden tachycardia, narrow QRS	5	128	84%	0.95	0.16
03_SPI2	sudden tachycardia, narrow QRS	4	105	78%	0.93	0.28

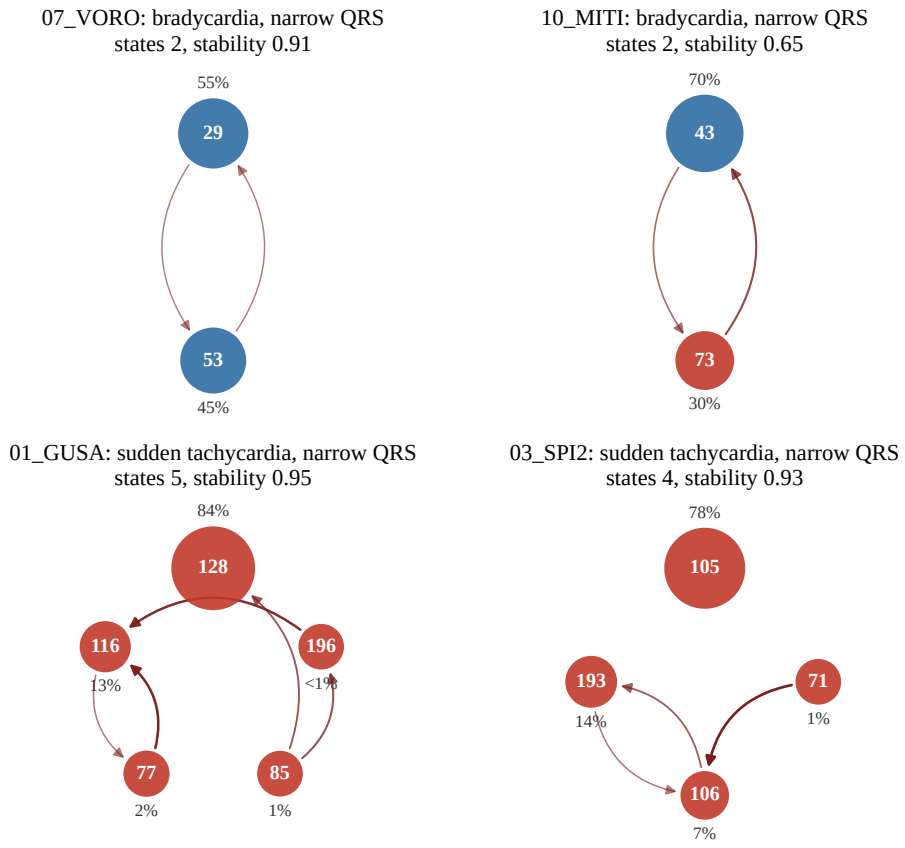


Figure S2: State graphs of the test-database recordings: two from the 'bradycardia, narrow QRS' section (top) and two from the 'sudden tachycardia, narrow QRS' section (bottom). The number in a vertex is the heart rate of the state (beats per minute), the percentage the share of time; arrows show transitions with a probability of at least 5%.